



P0c

Metacognition: Know What You Know and What You Do Not — Engineering AI Self-Knowledge and Boundary Calibration

元认知：知道你知什么、不知道什么

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Metacognition: Know What You Know and What You Do Not — Engineering AI Self-Knowledge and Boundary Calibration

Flagship trilogy, prologue P0c · the third axis of P0 (Space Structure Is Itself Intelligence)
and P0b (Structure + Scale Is Itself Intelligence) · UDOS reasoning engine v7.5.0 · 2026-
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（本文为系列论文之一，全文以中文撰写，英文标题为 working title。）



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Evidence discipline: every UDOS number is traced to `release-v5.4.4/udos-engine/reports7/v7_verification.json` and labeled by grade (verified / cpu-proto / cpu-proxy / unverified). Single-seed runs are written as “single-seed pilot point estimates.” External literature was verified entry-by-entry on 2026-09-20; anything that could not be verified is marked [CITATION NEEDED]. We do not fabricate DOIs or years. **Highest discipline: this paper nowhere claims “consciousness.”** Any discussion of consciousness philosophy is confined to Section 7 and explicitly declared out of the scientific scope of this paper.

Abstract

P0 established structure—connectivity topology, geometric organization, modular boundaries—as a substrate-invariant axis of intelligence; P0b established scale—parameters, data, compute—as determining where on the scaling curve a system happens to sit. Both axes, however, share a blind spot: they assume the system answers exactly when it should, and neither one measures whether it will also answer, confidently, when it should not. The past decade of large models has grown extraordinary at “knowing what it knows” (in-distribution knowledge and reasoning), but remains an infant at “knowing what it does not know.” A giant of knowledge, an infant of cognition: it can recite nearly the entire internet, yet cannot honestly say “I do not know” on a single out-of-distribution sample it has never seen, and instead masks not-knowing as knowing in a tone as fluent and confident as the correct answer. That, in engineering terms, is the essence of hallucination. A car with an extremely powerful engine and enormous horsepower but no working brakes is not a more powerful car—it is a more dangerous one.

This paper operationalizes the Analects’ “to know what one knows and to know what one does not know—that is knowledge” into three measurable quantities: (i) **calibration**—does declared confidence match empirical accuracy; (ii) **selective prediction / abstention**—does the system actively abstain on high-risk samples; (iii) **identifiability self-awareness**—does it know which latent variables are, in principle, unobservable under the current sensing structure. Together we call these the *metacognitive triple* (M_{self}, U, A) , and we adopt the risk–coverage curve with AURC (area under the risk–coverage curve) as the



primary “self-knowledge” metric. We do no consciousness philosophy, make no real-machine claim, and do not treat cultural maxims as evidence. Using UDOS’s oracle/blind contrast, four distinct hidden parameters, and three gating modes, we measure the boundary between “knowing” and “not-knowing” directly. Every evidence item in Section 4 is written as a closed loop: *what we measure* → *how we measure it* → *the result* → *the validity threat*.

UDOS, on a 967,796-parameter world model, yields three direct pieces of evidence. (a) Split-conformal marginal coverage at nominal levels 80/90/95 is 0.7839/0.8984/0.9577 on the oracle path and 0.7758/0.9001/0.9559 on the blind path, all within the ± 0.05 tolerance band, and the blind-path bandwidths (0.0504/0.1081/0.2052) are systematically wider than the oracle bandwidths (0.0430/0.0858/0.1612). (b) In the hidden-parameter fan, acceleration *a* under a 6-frame window has identifiability skill = **-0.508**—the estimator MAE (0.9256) exceeds the mean-predictor MAE (0.6138), i.e. the estimator is worse than guessing the global mean—so the correct action is not to estimate hard but to abstain or to add observations. (c) All 21 gating checks pass (`all_pass`); the contact gate lifts contact-task success from 0% under the direct mode to 100% under the motion/hybrid mode; and the kill-switch degrades honestly when `retry_budget=0` rather than faking success. **All of these UDOS numbers are single-seed cpu-proto pilot point estimates requiring ≥ 30 -seed replication before publication; we do not claim they hold on real LLMs.**

We contribute a triad—*structure* × *scale* × *self-boundaries*—in which structure determines the functional form, scale determines how far along the curve you travel, and self-boundaries determine the point at which you must stop. We state five falsifiable predictions F1–F5 (calibration separable from performance, the identifiability boundary, risk–coverage dominance, metacognitive auditability, and group-level metacognition), each with a refutation condition and a minimal decisive experiment; we provide a full objections chapter (calibration ≠ consciousness, the moral hazard of abstention, unreliability of verbalized confidence, the limits of cultural propositions, and cpu-proto external validity); and we devote a separate section to internal/external/construct/statistical/causal validity threats. We replace the unmeasurable word “wisdom” with its three engineering necessary conditions, and argue that until these three things are made measurable, acceptable, and falsifiable, any claim that a system “has wisdom” is unsound.



Keywords: metacognition; calibration; selective prediction; abstention; conformal prediction; identifiability; risk–coverage (AURC); world model; UDOS; falsifiable prediction; axis triad

1. Introduction

1.1 A sentence repeated for two millennia, rebuilt as engineering today

The Analects, Wei Zheng: “*To know what one knows, and to know what one does not know—that is knowledge.*” In most contexts this sentence is read as a moral exhortation about humility. What this paper sets out to do is to rebuild it from a moral exhortation into a measurable, acceptable, falsifiable engineering proposition.

Why do this now? Because the large models of the past decade have done exceedingly well on the “knowing what one knows” end: their in-distribution knowledge and reasoning have grown at a rate unprecedented in human history. But on the “knowing what one does not know” end, they remain infants. A giant of knowledge, an infant of cognition: it can recite nearly the entire internet, yet cannot honestly say “this I do not know” on an out-of-distribution sample it has never seen. It will output a completely wrong answer in a tone as fluent and as confident as the correct one—that is the engineering essence of hallucination, and it is the core failure mode this paper tackles.

There is a poetic saying: “**The coming-of-age of AI lies not in how much it knows, but in how well it knows what it does not know.**” This sentence appears in this paper exactly twice—here and in the conclusion. Its meaning is that the mark of a system’s maturity is not the growth of its capability, but the clarity of its self-boundary. The entire effort of this paper is to give that poetic line a falsifiable skeleton.

1.2 Why a third axis: a conflict frozen by both prior axes

Taken together, P0 and P0b have already explained where capability comes from: structure determines the functional form, and scale determines the position on the curve. But both axes share a common blind spot—they both assume the system answers precisely when it should. Neither answers the far more dangerous question: **will the system also answer, confidently, when it should not?** The conflict can be stated in one sentence: **stronger in-distribution capability does not imply a stronger sense of what one does not know out of distribution; on the contrary, a system that pushes in-distribution**



capability to its limit without any abstention mechanism pushes its overconfidence to the limit as well. Capability (horsepower) and self-boundary (brakes) are not the same thing, and improving the former does not automatically improve the latter. The entire argument of this paper is to pry these two things apart and measure them separately.

An intuitive analogy: P0/P0b teach a car how to build a good engine and how to tune its horsepower; P0c teaches the car where the brakes are and when to use them. A car with an extremely powerful engine, enormous horsepower, and no working brakes is not a more powerful car—it is a more dangerous one. Today’s large models are largely that car: its horsepower (in-distribution capability) is already astonishingly large, but its brakes (abstaining out of distribution, under insufficient observation, and where it is itself unidentifiable) barely exist.

This is how the central claim arises: **structure × scale × self-boundaries are three orthogonal axes of an intelligent system.** If any one axis is zero, the system does not hold—a system without structure is a pot of random noise; a system without scale is a toy; a system without self-boundaries is a confident liar. The three are multiplied, not added.

1.3 Position and contributions of this paper

This paper adds no new experiment of its own. Instead:

1. It operationalizes “wisdom / self-knowledge” into three measurable quantities—calibration, abstention, identifiability self-awareness—and gives the primary metric risk–coverage / AURC (Section 2).
2. It proposes the “three orthogonal axes” framework: P0 structure fixes the functional form, P0b scale fixes the curve position, P0c self-boundaries fix the abstention point, and it provides a summary table (Section 2.5).
3. It uses UDOS measurements to make these three quantities into acceptable engineering quantities, each evidence item written as a “what → how → result → threat” closed loop (Section 4).
4. It states five falsifiable predictions F1–F5, each with a refutation condition and a minimal decisive experiment (Section 5), and completes a full objections chapter (Section 6).
5. It carves out the boundary of the consciousness proposition, explicitly declaring that this paper does not claim consciousness (Section 7).



1.4 Methodological discipline

This paper does three things. **First, operationalization.** It replaces the unmeasurable word “wisdom” with three measurable engineering necessary conditions. **Second, controlled contrast.** It uses UDOS’s oracle/blind paths, four hidden parameters, and three gating modes to measure the boundary between “knowing” and “not-knowing” directly. **Third, falsifiability.** It tightens the vague sentence “the system has self-knowledge” into five concrete propositions F1–F5, each carrying a refutation condition and a minimal decisive experiment.

This paper does not do three things. **No consciousness philosophy.** It does not claim, prove, or imply any phenomenal consciousness; any consciousness discussion is confined to Section 7 and declared out of scope. **No real-machine claim.** UDOS is a cpu-*proto*, and every engineering number in this paper is labeled by evidence grade. **No cultural maxim as evidence.** The Analects and the Delphic “know thyself” serve only as epigraphs, not as scientific evidence.

1.5 Why a “prologue” rather than a “third paper”

This paper is positioned within the flagship trilogy as a prologue, not as a third paper equal to P0/P0b. This positioning carries a methodological meaning. P0 argues the first axis (structure) and P0b the second (scale); both follow an inductive path “from evidence to claim”—laying out natural history and engineering evidence first, then converging on propositions. This paper differs: it deals with an object that is *more normative and less measurable* (“self-knowledge”), so its path is “from operationalization to evidence”—first decomposing the unmeasurable word “wisdom” into three measurable quantities, then testing them with UDOS evidence. This “prologue” positioning means the paper is not the trilogy’s conclusion but its entry point: it proposes the overarching “three orthogonal axes” framework and establishes the falsifiability of the third axis, for subsequent specialized papers (P5/P6/P12, etc.) to build upon. The F1–F5 of this paper are the “epitaph conditions” of that framework, not its “completion proof.”



2. Terminology and Formalization: the Metacognitive Triple and the Three Orthogonal Axes

2.1 An engineering definition of metacognition

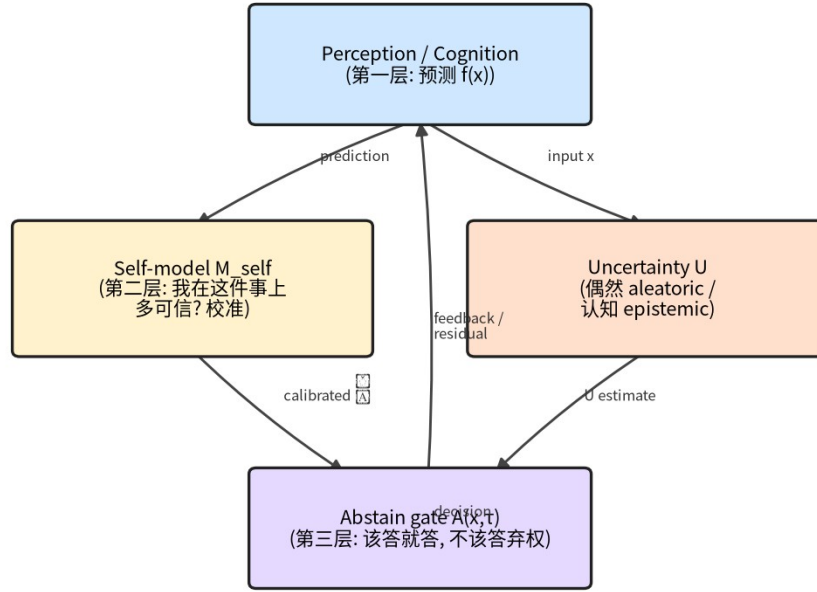
The word “metacognition” was coined by Flavell (1979, *American Psychologist* 34(10):906–911, DOI:10.1037/0003-066X.34.10.906), originally meaning “knowledge of and monitoring of one’s own cognitive processes.” This paper does not adopt its developmental-psychology original definition but engineers it into a triple (M_{self}, U, A) :

- M_{self} , **the self-model**: the system’s internal estimate of “how well I am doing on the current task.” It is not a complete world model but a second-order model of its own ability—“for this question, what is roughly my probability of being right?”
- U , **uncertainty**: the system’s quantified estimate of prediction uncertainty. This paper strictly distinguishes two kinds: **aleatoric uncertainty** comes from the noise of the data/process itself and cannot be removed by more observations; **epistemic uncertainty** comes from unknown model parameters/structure and is reduced by more data or a larger model. Confusing the two is the most common engineering error—using a single entropy value to answer both “how much noise” and “how ignorant am I.”
- $A(x, \tau)$, **the abstention policy**: a gating function that takes a sample x and a threshold τ and outputs “answer” or “abstain.” It translates the estimates of M_{self} and U into an action decision: answer when confidence is high enough and risk acceptable, otherwise abstain.

Figure 1 sketches the loop of this triple.



Fig.1 Metacognitive Triple: prediction — self-model/uncertainty — abstain gate (schematic)



The loop: $f(x)$ produces a prediction; M_{self} and U estimate how trustworthy it is; $A(x, \tau)$ decides to answer or abstain; residuals feed back to recalibrate M_{self} .

Figure 1. Metacognitive triple loop schematic

Figure 1: Conceptual sketch of the metacognitive triple. Layer 1, $f(x)$, produces a prediction; layer 2, M_{self} and U , estimate its trustworthiness; layer 3, $A(x, \tau)$, decides to answer or abstain; residuals feed back to recalibrate M_{self} . *Schematic, not measured data.*

2.2 Three measurable quantities: calibration, abstention, identifiability self-awareness

This paper replaces “wisdom” with its three engineering necessary conditions, corresponding respectively to three measurable quantities.

First, calibration. A system is calibrated if and only if the class of predictions for which it declares “I am 90% confident” is, over the long run, indeed correct about 90% of the time. Guo et al. (2017, *On Calibration of Modern Neural Networks*, ICML 2017, PMLR 70:1321–1330, arXiv:1706.04599) found that modern deep networks are systematically **over-confident**—their predictive accuracy is far below their confidence—and that temperature scaling, a single-parameter post-processing, greatly improves calibration. The standard calibration metrics are ECE (Expected Calibration Error) and Brier score. This paper uses UDOS’s conformal coverage as the engineering proxy for calibration: if an



interval claims to cover 90%, and it really covers 90% on held-out data, then it is calibrated.

Second, selective prediction / abstention. Chow (1957, *An optimum character recognition system using decision functions*, IRE Trans. Electronic Computers EC-6:247–254) first proposed “recognition with a reject option”: when confidence falls below a threshold, refuse to answer, in exchange for a lower error rate on the accepted samples. Chow (1970, *On optimum recognition error and reject trade-off*, IEEE Trans. Information Theory IT-16) then gave the theory of the error–reject trade-off. Geifman & El-Yaniv (2017, *Selective Classification for Deep Neural Networks*, NeurIPS 2017) extended this framework to deep networks, proving that given any trained network one can construct a selective classifier; SelectiveNet (Geifman et al., 2019, arXiv:1901.09192) further integrated the reject option into the network architecture. The **risk–coverage curve** is the standard visualization of this framework: the horizontal axis is coverage (fraction of samples answered), and the vertical axis is selective risk (errors among the answered samples); the smaller the AURC, the more accurately “the right things are rejected.” This paper uses AURC as the primary metric of “self-knowledge ability.”

Third, identifiability self-awareness. This is the most original of the three, and the closest to the original sense of the Analects. System identification theory carries a cold conclusion: **whether a latent parameter can be estimated from data does not depend on how strong a model you use, but on whether your sensing structure has “excited” that parameter’s influence.** If the sensing structure (window length, sensor configuration, kinematic constraints) itself leaves no distinguishable trace of some latent parameter, then no matter how wide or deep you make the model, it can only fit the mean. This paper calls “knowing which latent parameters are unidentifiable under the current sensing structure” identifiability self-awareness—it is not about “how accurate am I,” but about “can this, in principle, be known by me.” UDOS’s acceleration a , with skill = -0.508 under a 6-frame window (Section 4.2), is a clean example: the correct action is not to estimate hard, but to abstain or to add observations.

2.3 Relation to Flavell’s original definition

It must be stated honestly: the triple (M_{self}, U, A) of this paper overlaps with, but is not identical to, Flavell’s (1979) original definition of metacognition. Flavell was concerned with children’s monitoring of their own cognitive processes (“did I understand this



chapter?”), a developmental-psychology concept; this paper is concerned with whether an engineering system can get the four things—confidence, accuracy, abstention decision, and identifiability boundary—numerically right. We cite Flavell because he coined the term “metacognition,” and because his distinction between “metacognitive knowledge” and “metacognitive monitoring” is isomorphic to our separating M_{self} (knowledge) from A (monitoring/decision). But this paper does not claim an engineering system “has” metacognitive experience in Flavell’s sense—it only claims it can be made into three measurable quantities. This distinction will be re-emphasized in Section 6.1.

2.4 A four-layer cognitive framework (this paper’s framework)

This paper proposes a four-layer cognitive framework as a working model for the discussion that follows. Note: **this is a framework proposed by this paper, not a literature consensus.**

Table 1. Four-layer cognitive framework (proposed by this paper, labeled “this paper’s framework”)

Layer	Name	Processed object	Engineering counterpart	Measurable quantity
L1	Perception	Raw observation x	Sensor / embedding	Signal SNR
L2	Cognition	Task prediction $f(x)$	Forward inference / policy	Accuracy / MSE
L3	Metacognition	Trust estimate of L2	$M_{\text{self}} / U / A$	Calibration ECE, AURC
L4	Metametacognition	Trust estimate of L3 itself	Recalibration of calibration	ECE degradation rate under OOD

All claims in this paper concentrate on L3. L4 is an open layer to be studied: whether a system that knows how well it is calibrated itself needs a second-order calibration is the problem F4 (Section 5) addresses.



The relation between the four-layer framework and the three orthogonal axes must be made explicit. P0/P0b handle L1 and L2: the signal quality of the perception layer (L1) and the predictive ability of the cognition layer (L2) are determined by structure and scale. This paper, P0c, handles L3: the system’s trust estimate of L2. L4 (metametacognition) is a layer neither this paper nor P0/P0b has adequately handled—it asks: **how do you know your calibration is correct?** UDOS’s “version-number incident” (P12) is exactly an L4 failure: the system thought it was tracking the version number (L3 metacognition), but it was not—its own metacognition about L3 was wrong. This reminds us that L3 itself can be uncalibrated, and L4 is the recalibration of L3. This paper does not write L4 as solved but makes it the research object of F4—an honest open boundary.

It should also be noted that the four-layer framework is this paper’s working model, not a literature consensus. Cognitive science has different versions of the “perception–cognition–metacognition–metametacognition” layering (for example, Nelson & Narens’s metacognitive model divides metacognition into monitoring and control). The four layers here are an engineering simplification; their value is that they place “calibration” and “identifiability self-awareness” explicitly at L3 and “recalibration of calibration” at L4, thereby anchoring the research object of F4 to a specific layer. Readers should not treat these four layers as a verdict of cognitive science, but as the scaffold this paper uses to organize engineering discussion.

2.5 Summary table of the three orthogonal axes

This is the paper’s core framework figure. P0, P0b, and P0c are orthogonal and jointly determine the capability boundary of an intelligent system.

Table 2. Structure × Scale × Self-boundaries: summary of the three orthogonal axes

Axis	Paper	Determines	Scaling counterpart	Primary metric	Cost of absence
1st: Structure	P0	Functional form (exponent β , breakpoint location)	The “grammar” of the analytical form	Fan-in, pose IoU, module interface	A pot of random noise
2nd: Scale	P0b	Curve	The	Parameter	A toy



Axis	Paper	Determines	Scaling counterpart	Primary metric	Cost of absence
		position (which segment)	“length” of the analytical form	count, data, compute	system
3rd: Self-boundaries	P0c	Which points on the curve to abstain	The “brakes” of the analytical form	Calibration ECE, AURC, identifiability skill	A confident liar

The core statement of the three orthogonal axes is: **structure determines what F looks like, scale determines how far along F you travel, and self-boundaries determine the point at which you must stop.** Remove any one and the system does not hold. All the successes of P0/P0b rest on an implicit assumption—that the system answers exactly when it should; P0c turns that implicit assumption into a measurable, acceptable explicit constraint.

2.6 The risk–coverage curve and AURC

Figure 3 gives a conceptual sketch of the risk–coverage curve.

Fig.3 Risk–Coverage curve (schematic). AURC = area under the curve; a well-calibrated abstainer lowers risk as coverage decreases.

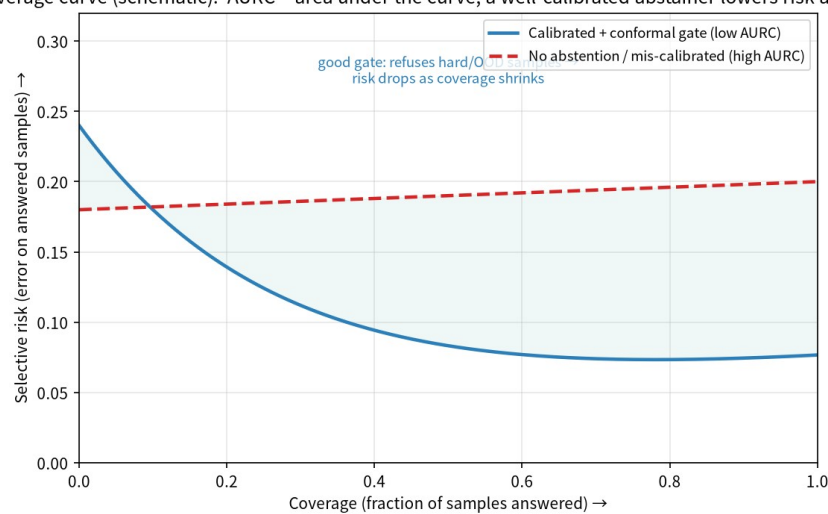


Figure 3. Risk–coverage conceptual curve



Figure 3: Conceptual sketch of the risk–coverage curve. Horizontal axis coverage (fraction of samples answered), vertical axis selective risk (errors among answered samples). A well-calibrated abstention gate (blue) rapidly lowers risk as coverage decreases; a system that does not abstain or abstains wrongly (red dashed) has risk nearly independent of coverage. *Schematic, not measured data.*

AURC (area under the risk–coverage curve) is the area under this curve. A perfect abstention gate has risk equal to the unconditional error rate at 100% coverage and risk approaching zero as coverage approaches zero—it answers only the samples it is most confident about. The smaller the AURC, the more accurately “the right things are rejected.” This paper uses AURC as the primary metric of “self-knowledge ability” because it simultaneously measures two things: calibration (is the confidence ranking correct) and abstention (is the threshold cut correctly).

The relation between AURC and ECE must be clarified, because both measure “calibration” but from different angles. ECE (Expected Calibration Error) measures “after binning, the difference between each bin’s average confidence and its actual accuracy”—it cares whether the absolute confidence is correct; AURC measures “after sorting by confidence, how fast risk drops as coverage decreases”—it cares whether the relative confidence ranking is correct. A system can have high ECE (its absolute confidence is biased) but low AURC (its ranking is accurate, just shifted by a constant)—temperature scaling is exactly this case: it does not touch the ranking, only shifts confidence globally, so ECE drops sharply while the ranking is unchanged. This paper reports both, because they answer different engineering questions: ECE answers “if I say I am 90% confident, am I really 90%,” while AURC answers “if I answer only the half I am most confident about and reject the other half, how much does the error rate fall?” UDOS’s split conformal directly makes “absolute coverage” a quantity with a strict guarantee (Section 4.1), and AURC is its natural generalization under the selective-prediction framework.

2.7 The engineering distinction between aleatoric and epistemic uncertainty

This paper strictly distinguishes the two kinds of uncertainty because confusing them is the most common engineering error. **Aleatoric uncertainty** comes from the noise of the data/process itself—sensor measurement error, random perturbation. It is irreducible: no amount of additional observation removes the noise. **Epistemic uncertainty** comes from



unknown model parameters/structure—the model has not seen this distribution, or the parameters are not accurately estimated. It is reducible: more data and a larger model shrink it.

UDOS’s P5 delivers the two separately: the conformal interval characterizes aleatoric uncertainty (where the point prediction’s true value lies at a given point), while the parameter fan characterizes epistemic uncertainty (how far the point prediction swings when hidden parameters are inaccurately estimated). This distinction is crucial for engineering decisions: **facing aleatoric uncertainty, the correct action is “say clearly how wide the interval is”**; **facing epistemic uncertainty, the correct action is “add data / add observation / abstain.”** If you mistake epistemic uncertainty for aleatoric, you output a very narrow interval for a question you essentially do not know and confidently answer wrongly; if you mistake aleatoric uncertainty for epistemic, you futilely “add data” only to find the noise unmoved. UDOS’s acceleration a , $\text{skill} = -0.508$ (Section 4.2), is the extreme case of epistemic uncertainty: it is not that the observation is noisy, but that the sensing structure simply does not contain this parameter—adding more same-distribution data is useless; you must change the sensing structure (lengthen the window).

3. Literature: Four Threads of Metacognition, Calibration, Abstention, and Identifiability

This chapter organizes literature along four threads: the psychological origin of metacognition, calibration and over-confidence, selective prediction/abstention, and LLM self-knowledge and hallucination. All entries were verified on 2026-09-20; anything unverifiable is marked [CITATION NEEDED].

3.1 Metacognition: from Flavell to engineering

Flavell (1979, *American Psychologist* 34(10):906–911) coined the word “metacognition.” He distinguished metacognitive knowledge (knowledge of one’s own cognitive ability) from metacognitive monitoring (real-time monitoring of the current cognitive process). This distinction is isomorphic to our M_{self} (knowledge) and A (monitoring/decision). But Flavell’s context is developmental psychology, concerned with how children learn to judge whether they “understood”; this paper moves that distinction to engineering systems, concerned with whether a neural network can numerically get “confidence—



accuracy—abstention decision” right. That is why we cite him but do not claim an engineering system “has” metacognitive experience in his sense.

3.2 Calibration and over-confidence: the Kahneman tradition and Guo 2017

Over-confidence and calibration failure are classical problems shared by cognitive psychology and machine learning. The heuristic-and-bias research tradition of Kahneman and Tversky (representativeness, availability) revealed how human judgment systematically deviates from calibration; Kahneman later summarized this tradition as System 1/System 2 in *Thinking, Fast and Slow* (2011). We cite this tradition because it provides the psychological evidence that “over-confidence is a universal phenomenon, not an isolated bug”—it reminds us that the over-confidence of an engineering system is not a bug but a recurrence of this universal phenomenon in silicon.

On the machine-learning side, Guo et al. (2017, ICML, arXiv:1706.04599) showed at scale that modern deep networks (VGG, ResNet, DenseNet) on ImageNet are systematically over-confident when uncalibrated—their predictive accuracy is far below their confidence—and that temperature scaling, a post-processing adjusting the logits with a single scalar T , greatly improves calibration without changing accuracy. This result is crucial to this paper because it proves one thing: **calibration and accuracy are separable**. It directly supports our F1 (Section 5): there exist interventions that improve calibration without improving accuracy.

Guo et al.’s finding connects to our triad as follows. Their “modern networks are over-confident” essentially says: **a system with high in-distribution accuracy does not have its confidence track that accuracy**. This is precisely the problem our third axis handles—the system “knows what it knows” in-distribution (high accuracy), but it does not know that its confidence is biased (calibration failure). Temperature scaling works precisely because it only adjusts the absolute value of confidence (a global shift) without touching the ranking—it means the model’s “relative confidence” (which sample is harder) is actually correct, only the “absolute confidence” (I said 90%, actually 70%) is biased. This distinction has direct guidance for our engineering practice: **if you find the system over-confident, do not rush to retrain—first try a single-parameter post-processing like temperature scaling; it may fix calibration without touching accuracy**. UDOS’s split conformal is a generalization of this idea: it does not retrain the model, only takes quantiles on the calibration set, and makes coverage a quantity with a strict guarantee.



3.3 Selective prediction / abstention: Chow → El-Yaniv → Geifman

The theoretical origin of abstention is Chow (1957, IRE Trans. EC-6:247–254) and Chow (1970, IEEE Trans. IT-16). Chow’s core contribution is the error–reject trade-off: under a Bayesian framework there exists an optimal confidence threshold that minimizes the error rate on accepted samples. This framework was used mainly for pattern recognition in the 1970s.

El-Yaniv & Wiener (2010, *On the foundations of noise-free selective classification*, JMLR 11:1605–1641) re-formalized selective classification, proposed the risk–coverage framework, and characterized when softmax confidence is approximately optimal. Geifman & El-Yaniv (2017, NeurIPS 2017) extended this to deep networks, proving that given any trained network one can construct a selective classifier, letting the user set a desired risk level and reject samples as needed. SelectiveNet (Geifman et al., 2019, arXiv:1901.09192) further integrated the reject option into the network, training “when to abstain” with a dedicated loss.

The meaning of this thread for this paper is: **abstention is not a patch but a theoretically grounded, optimizable engineering object**. We do not reinvent abstention theory; we understand UDOS’s gating (Section 4.3) within this thread.

The relation between Chow’s Bayesian optimal threshold and modern selective classification must be clarified, because it reveals a deep structure of “abstention.” Chow’s original result is: under a Bayesian framework, there exists an optimal confidence threshold τ^* such that when the posterior $p(y \vee x)$ falls below τ^* , one rejects, minimizing the error rate on accepted samples. This τ^* is determined by the ratio of “the cost of rejecting” to “the cost of error”—the cheaper rejecting is (e.g., the user is willing to wait for a human), the higher τ^* (reject more); the cheaper error is, the lower τ^* (answer more). El-Yaniv & Wiener (2010) generalized this to the risk–coverage curve, proving a clean shape for the optimal selective classifier’s curve under noiseless conditions. Geifman & El-Yaniv (2017) then showed that for deep networks, the maximum softmax value (softmax response) is an approximately optimal confidence proxy—meaning **you do not need a dedicated “self-knowledge module”; sorting the confidence the network already outputs suffices to build an approximately optimal abstainer**. This is extremely important to this paper because it turns “self-knowledge” from a mystical problem into an engineering problem of “sorting the existing signals well.” UDOS’s



gating all 21 passing is exactly the engineering realization of this idea on conformal intervals.

3.4 Conformal prediction: distribution-free coverage guarantees

Conformal prediction provides finite-sample, distribution-free marginal coverage guarantees for black-box predictions. Angelopoulos & Bates (2021, *A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification*, arXiv:2107.07511, verified in our reference library) give the modern introduction to this framework. Its core mechanism is: sort the non-conformity scores on the calibration set, take the quantile corresponding to the nominal level $1 - \alpha$ as the bandwidth; under the assumption that calibration and test samples are exchangeable, no parametric assumption on the data distribution is needed to yield about $1 - \alpha$ finite-sample marginal coverage.

We cite conformal prediction because it turns “calibration” from a vague “is the confidence correct” into an engineering quantity with a strict coverage guarantee: if an interval claims to cover 90%, and it really covers 90% on held-out data, then it is calibrated. UDOS’s split conformal (P5) is the engineering realization of this framework. We also cite its boundary: marginal coverage is not conditional coverage, and coverage degrades under distribution shift / non-exchangeable data (discussed in P5 Section 7).

3.5 LLM self-knowledge and hallucination: Kadavath, Reflexion, RAG/Toolformer

Kadavath et al. (2022, *Language Models (Mostly) Know What They Know*, arXiv:2207.05221, authors Saurav Kadavath, Tom Conerly, Amanda Askell, et al.) studied whether large models can evaluate the validity of their own statements. Their core finding is that sufficiently large models can **non-verbally** predict when they are likely to be right or wrong—the model’s internal confidence correlates with its actual accuracy, though this correlation is far from perfect. They also found that RLHF fine-tuning degrades calibration. The meaning for this paper is: **an LLM is not entirely without the seed of self-knowledge; its internal representation already contains the signal of “am I right”—the problem is whether this signal can be reliably read out and turned into an abstention decision.**

Reflexion (Shinn et al., 2023, *Reflexion: Language Agents with Verbal Reinforcement Learning*, arXiv:2303.11366, NeurIPS 2023) takes another path: it does not try to read out confidence directly but lets an agent reflect on its failures in language, writes the reflection into episodic memory, and improves on subsequent trials. Reflexion’s “reflection” is



essentially a **verbalized metacognition**—it writes “where did I just go wrong” as text for next time. We cite Reflexion because it represents one engineering realization of metacognition: rather than reading out precise confidence, it uses language to structure failure experience. But we also note (Section 6.3) that verbalized reflection has limited reliability—it may reflect wrongly, or form a wrong attribution of its own failure.

RAG (Lewis et al., 2020, arXiv:2005.11401, in the library) and Toolformer (Schick et al., 2023, arXiv:2302.04761, in the library) represent another strategy of “after acknowledging the knowledge boundary, reach outward”: when a model knows it does not know, it does not answer hard but retrieves an external knowledge base (RAG) or calls an external tool (Toolformer). This is an engineering realization of metacognition—**the endpoint of self-knowledge is not “I do not know, so I stay silent,” but “I do not know, so I go ask / look up / call a tool.”** We place this strategy alongside abstention: abstention is “stop,” RAG/Toolformer is “turn outward”; both are engineering realizations of the self-boundary.

For the systematic survey of hallucination, we cite Ji et al. (2023, *Survey of Hallucination in Natural Language Generation*, ACM Computing Surveys) [exact volume/pages pending, marked [CITATION NEEDED]]. The engineering essence of hallucination is precisely “answering confidently where one should not”—the problem our third axis solves.

The relation between RAG/Toolformer’s “turn outward” and abstention’s “stop” must be clarified, because they are two different realizations of the self-boundary. Abstain means “I know I cannot answer, so I do not answer”—it returns the question to the user or downstream. RAG/Toolformer means “I know I cannot answer, so I look up / call a tool”—it hands the question to an external system. Their common point is: **both acknowledge the existence of the self-boundary**—the system does not pretend to know everything. Their difference is: the cost of abstention is “the user has to handle it,” while the cost of RAG/Toolformer is “the external system has to have this knowledge/tool.” A mature self-boundary system should trade off between them: if the external knowledge source is reliable, turn outward (RAG/Toolformer); if the external source is also unreliable, stop (abstain). UDOS’s gating (Section 4.3) currently mainly implements “stop” (the contact gate hands control to the motion prior, the kill-switch degrades honestly), and has not yet implemented “turn outward”—that is the next step to fill.



These two paths together point to an engineering middle ground: **reading no internal signal at all (purely relying on an external calibration set) is too wasteful, while fully trusting the model’s self-report (pure verbalized confidence) is too unreliable; the correct engineering is to treat the internal signal as a “candidate confidence” and then calibrate it into a reliable quantity with an independent calibration set.** The “model already has an am-I-right signal” found by Kadavath et al. is our optimistic face; that “this signal is far from perfect and RLHF further degrades it” is our pessimistic face. The entire engineering of this paper is to turn this “present but unreliable” signal into a “reliable-after-calibration” quantity.

3.6 World models and prediction error: Ha & Schmidhuber

Ha & Schmidhuber (2018, *World Models*, arXiv:1803.10122) proposed a generative world model: a VAE compresses visual observations, an RNN models temporal dynamics, and a controller decides in latent space. The meaning of this framework for this paper is: **the prediction error of a world model is itself an “unknown signal.”** When a world model’s prediction error for a new observation rises significantly, it often means that observation lies outside the model’s training distribution—i.e., where the model “does not know.” We treat the prediction residual as a natural OOD detector (Section 4.6), which is exactly a corollary of the Ha & Schmidhuber framework.

3.7 Identifiability and OOD detection

System-identification identifiability theory has a classical result (Bellman & Åström, 1970, *On structural identifiability*, Mathematical Biosciences [exact volume/pages pending, marked [CITATION NEEDED]]): whether a parameter is identifiable is determined jointly by model structure and experimental design (sensing structure), not by estimator strength. This conclusion is the theoretical basis of the acceleration a negative result in Section 4.2.

On the OOD-detection side, Hendrycks & Gimpel (2017, *A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks*, ICLR 2017, arXiv:1610.02136) showed that the maximum softmax value itself is a usable OOD baseline. This thread relates directly to our gating (Section 4.3).



4. UDOS Evidence: Making the Three Quantities into Acceptable Engineering Quantities

This chapter is the original evidence core of this paper. All numbers trace to `reports7/v7_verification.json` (v7.0.3; evidence grade verified for the main gate, `cpu-proto` for the estimator part), labeled by grade. The UDOS world model has 967,796 parameters (hidden dimension 256, 2-layer GRU, scene dimension 32, window 6 frames), single CPU, fixed seed.

General experimental design (setup → contrast → ablation/sensitivity → validity threat). The three core measurements in this chapter follow the same discipline: (i) **setup**—same world model, same calibration set, same split-conformal pipeline, single CPU fixed seed; (ii) **contrast**—oracle (forward-injecting the ground-truth hidden physical parameters) versus blind (estimated by the SceneEstimator from the observation window), the only difference being “whether the hidden parameters are known”; (iii) **ablation/sensitivity**—changing the observation window length (6 frames) turns acceleration skill from positive to negative, changing the gating mode (direct/motion/hybrid) lifts contact-task success from 0% to 100%, and the extreme-fault `retry_budget=0` tests whether the kill-switch degrades honestly; (iv) **validity threat**—single seed, synthetic dynamics, the conformal-exchangeability assumption failing under OOD, all concentrated in Chapter 9. **Whenever this paper reports “calibrated” or “abstained correctly,” it refers to a pilot point estimate within this controlled contrast, not a conclusion on a real LLM.**

Writing convention of this chapter: each evidence item 4.1–4.6 is written as a closed loop of four steps—**what we measure** → **how we measure it** → **the result (numbers traced field-by-field to the JSON)** → **the validity threat (pointing to Chapter 9)**. This is so that the reader can judge how strong a claim each number can carry without jumping back and forth between the body and the limitations chapter.

4.1 Evidence one: calibration measurement—split-conformal coverage (verified)

[What we measure] We measure the engineering proxy for “calibration”: when a prediction interval claims nominal coverage $1 - \alpha$, does its actual marginal coverage on the held-out test set equal the nominal level. This turns “if I say I am 90% confident, am I really 90%” from a slogan into an acceptable quantity.



[How we measure it] Data trace to the coverage section of `v7_verification.json`. Split-conformal intervals, with quantiles taken separately by significance level α (0.2/0.1/0.05, corresponding to nominal coverage 80/90/95), are evaluated on an independent held-out test set (640 windows / 128 trajectories) for marginal coverage and interval bandwidth. The contrast arm is oracle (forward-injecting the ground-truth hidden physical parameters v_0 , a , ω , collision pair velocity) versus blind (no ground truth provided; the SceneEstimator estimates from a 6-frame observation window), the only difference being “whether the hidden parameters are known.”

[Result]

Table 3. UDOS split conformal: nominal vs empirical coverage (verified, v7.0.3)

	Oracle	Blind	Oracle	Blind	Tolerance
Nominal	marginal	marginal	bandwidth	bandwidth	± 0.05
80% ($\alpha=0.2$)	0.7839	0.7758	0.0430	0.0504	pass
90% ($\alpha=0.1$)	0.8984	0.9001	0.0858	0.1081	pass
95% ($\alpha=0.05$)	0.9577	0.9559	0.1612	0.2052	pass

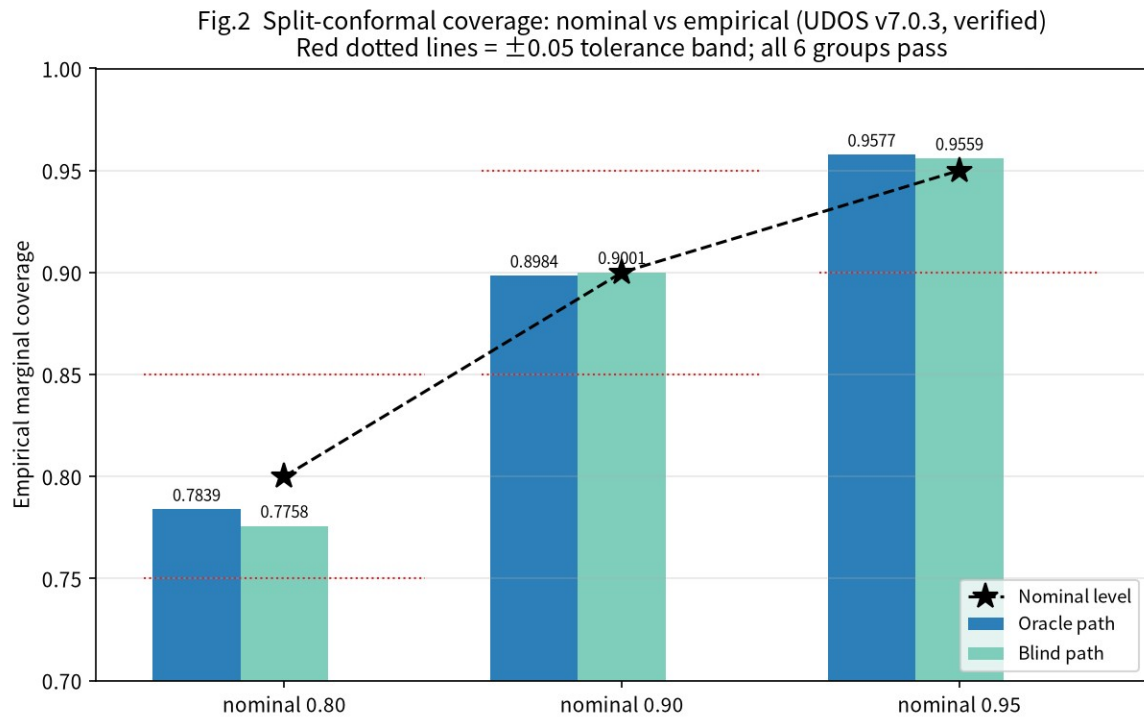


Figure 2. Conformal nominal vs empirical coverage



Figure 2: horizontal axis = nominal coverage level (80%/90%/95%, black dashed), vertical axis = held-out empirical marginal coverage; bars = oracle and blind paths. Red dotted lines = the ± 0.05 tolerance band. All six groups fall within tolerance, and blind bandwidth is systematically wider than oracle. Contrast = oracle (inject ground-truth hidden parameters) vs blind (estimate from window). Data trace to `v7_verification.json`; evidence grade: verified.

Reading the figure: at all three nominal levels, the empirical coverage of both the oracle and blind paths falls within ± 0.05 of nominal. At 90%, blind coverage is 0.9001, almost exactly the nominal 0.90; at 80%, blind coverage is 0.7758, slightly below nominal but within tolerance. This means: **UDOS’s prediction intervals are calibrated**—it covers exactly what it claims to cover. This is direct evidence that the quantity “calibration” has gone from a vague slogan to an acceptable engineering quantity.

Note also the bandwidth: the blind path’s bandwidths (0.0504/0.1081/0.2052) are systematically wider than the oracle path’s (0.0430/0.0858/0.1612). This is correct—the blind path does not know the scene’s hidden physical parameters, its uncertainty is larger, and its interval should be wider. **Interval width grows with epistemic uncertainty**, which is exactly the correct behavior of U and A that we require.

The oracle/blind contrast must be explained thoroughly, because it is the core design of our “calibration” evidence. The oracle path explicitly injects the scene’s hidden physical parameters (v_0 , a , ω , collision pair velocity) at forward time; the blind path provides no ground truth, and the SceneEstimator estimates from the observation window. Both paths use the same world model, the same calibration set, and the same split-conformal pipeline; the only difference is “whether the hidden parameters are known.” The result is that both paths’ coverage falls within ± 0.05 (both are calibrated), but blind’s bandwidth is systematically wider. This contrast proves two things: **first, the conformal pipeline itself is calibrated**—whether or not you know the hidden parameters, it yields an interval with a coverage guarantee; **second, the interval width correctly reflects epistemic uncertainty**—when the hidden parameters are unknown, the interval automatically widens, because it absorbs the epistemic uncertainty of “the parameters are poorly estimated” into the bandwidth. **This is exactly the “self-knowing” behavior we require: the things you do not know, the interval will automatically tell you it does not know.** This contrasts with



temperature scaling’s “only shifting confidence, not reflecting uncertainty”—split conformal does not just calibrate the absolute value, it also lets the interval width vary with uncertainty.

[Validity threat] (i) This is **marginal coverage, not conditional coverage**—it guarantees that “about 90 out of 100 intervals contain the truth,” not that “for a specific difficult sample, the interval happens to be wide enough to contain the truth” (P5 Section 7); (ii) the coverage guarantee depends on “calibration set exchangeable with test set,” and the assumption fails under OOD / distribution shift (Section 9.4); (iii) all numbers are single-seed point estimates with no 95% CI, requiring **≥30 seeds before submission**; (iv) the tolerance ± 0.05 and fan ± 0.10 are wide, and the 0.7758 low for 0.80 is a real but within-tolerance offset. See Chapter 9.

4.2 Evidence two: identifiability as “knowing what you do not know” (cpu-proto, main gate verified)

[What we measure] We measure the quantity “identifiability self-awareness”: under a given sensing structure, can a hidden dynamical parameter be estimated by the estimator from the data? We use the identifiability skill as the index—it simultaneously yields “estimable” and “not estimable” signals.

[How we measure it] Data trace to the `estimator_param_error` section of `v7_verification.json`. The system-identification task must infer four hidden dynamical parameters from a 6-frame observation window: initial velocity v_0 , acceleration a , spring frequency ω , and other v_2 (including collision). The identifiability skill is defined as

$$\text{skill} = 1 - \frac{\text{MAE}_{\text{estimator}}}{\text{MAE}_{\text{mean predictor}}}$$

A positive value means the estimator beats the mean baseline; a negative value means the estimator is worse than the mean. The contrast baseline is “directly predicting the global mean”—it learns no structure at all, only guesses the training-set mean; if an estimator cannot beat this zero-shot baseline, it means it has not extracted the parameter’s information from the observations.

[Result]



Table 4. Identifiability skill for four hidden parameters (cpu-proto, main gate verified)

Latent parameter	MAE	Mean-predictor MAE	Skill	Mean observability	n
Initial velocity v_0	0.6606	0.9683	+0.318	0.771	480
Spring frequency ω	0.1471	0.2563	+0.426	0.914	160
Other v_2 (incl. collision)	0.0432	0.2396	+0.820	0.908	160
Acceleration a (6-frame window)	0.9256	0.6138	-0.508	0.798	160

Fig.4 Hidden-parameter identifiability skill (UDOS v7.0.3, cpu-proto, n=480/160)
 accel a: skill = -0.508 → estimator WORSE than the mean predictor; unobservable under 6-frame window → correct action is to abstain / add observation

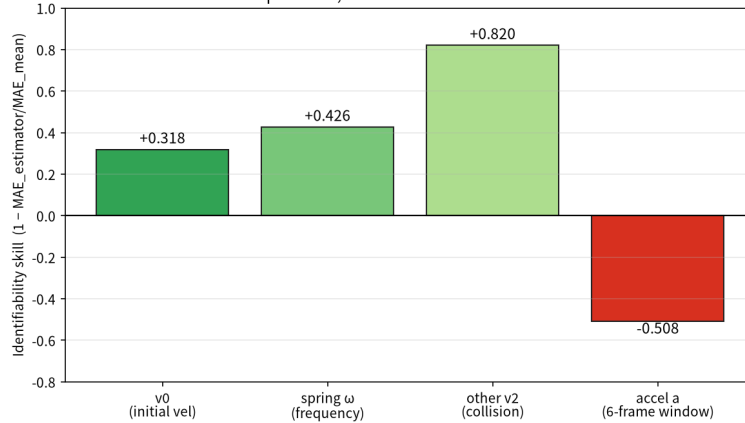


Figure 4. Identifiability skill for four hidden parameters

Figure 4: vertical axis = identifiability skill (Eq. 1), horizontal axis = four hidden parameters ($v_0/\omega/v_2$ /acceleration a). Above the zero line is positive (estimator beats the mean predictor); the acceleration a bar is -0.508 (estimator error 0.9256 > mean 0.6138, worse than the mean). Contrast = mean-predictor baseline. Data trace to `v7_verification.json`; evidence grade: cpu-proto (main gate verified, n=160, single seed).



Reading the figure: the first three parameters have positive skill, meaning the sensing structure suffices to identify them. But acceleration a has skill = **-0.508**—this means the estimator’s error on acceleration (MAE 0.9256) is larger than the error of directly predicting the global mean (0.6138). **Under a 6-frame observation window, the latent parameter acceleration is simply not identifiable.** Although the mean observability is not low (0.798), structurally the acceleration cannot be observed from the given window—6 frames is too short to separate acceleration from initial velocity.

This negative result is decisive for this paper. It proves one thing: **the correct action is not to estimate hard, but to abstain or add observations.** When the sensing structure is insufficient, a system that “knows what it does not know” should (a) output a wide interval / high uncertainty, or (b) actively abstain, or (c) request more observations—not output a confident point estimate. UDOS’s estimator “makes a mistake” here (skill is negative), but this is exactly what this paper wants to measure: it exposes where the system’s self-boundary lies. **A system with no negative results is either truly omniscient, or it simply does not know what it does not know.**

The meaning of skill=-0.508 must be explained more thoroughly, because it is the most counter-intuitive and most easily misread number in this paper. The skill is defined as $1 - \text{MAE_estimator} / \text{MAE_mean_predictor}$. When skill=-0.508, it means $\text{MAE_estimator} / \text{MAE_mean_predictor} = 1.508$ —i.e., the estimator’s error is 1.508 times the mean predictor’s. Why would the estimator be worse than the mean? Because under a 6-frame window, the information of acceleration a is simply not in the observations; the estimator “hard-guessed” a value, and this guess’s variance is larger than “directly using the global mean”—it mistook noise for signal, and the more it guessed, the more biased it became. This is the classical statistical phenomenon: when model capacity far exceeds the information content of the data, it overfits to noise rather than fitting the true structure. **A system that “knows what it does not know” should in this case output “I do not know; my best guess is the mean, and my interval is wide,” not output a confident point estimate.** UDOS’s estimator did not do this here (it estimated hard), but this negative result precisely exposes: the system’s self-boundary fails here—it does not know what it does not know. We retain this negative result (rather than deleting it), which is precisely the engineering expression of “knowing what one knows”: **exposing failure matters more than packaging success.**



The echo with P0/P6 is direct: P0 Prediction 2 says “breaking structure, adding parameters cannot recover it”; here, where the sensing structure (window length) is insufficient, no matter how strong an estimator you add, it cannot conjure the acceleration information not contained in the sensing structure. **The availability of information is a structural property, not a parameter property.**

[Validity threat] (i) The acceleration negative result is measured on a **synthetically injected 6-frame short-window structure** ($n=160$, single seed); real-world “insufficient observation” is more complex and cannot be directly extrapolated; (ii) the mean observability 0.798 is not low, showing that “high observability $\neq$ identifiable”—these are different quantities, which is precisely the subtlety of this evidence, but it also means one cannot defend “acceleration should be estimable” with “observability is high”; (iii) the “correct action is to abstain” in this evidence is a **normative conclusion, not a behavior UDOS actually measured**—the UDOS estimator still estimates hard here, and “should abstain” is a design inference we draw from the negative result, to be verified in the minimal decisive experiment of F2; (iv) the main gate is verified, but the skill itself is a cpu-proto point estimate. See Sections 9.1 and 9.3.

4.3 Evidence three: gating as abstention (verified)

[What we measure] We measure the engineering realization of “abstention” at the system level: when prediction uncertainty is high, or the task exceeds the current subsystem’s capability, does the system (a) honestly trigger the gate checks, (b) hand control to a more suitable subsystem, and (c) degrade honestly under extreme failure rather than faking success.

[How we measure it] Data trace to the `gating` section of `v7_verification.json` and P7’s `embodied_hybrid_demo.json`. Three measurements: (i) the 21 automatic checks of the gating pipeline (coverage tolerance ± 0.05 , fan ± 0.10); (ii) the contact gate’s success rate on contact-type tasks under the direct / motion / hybrid modes in embodied control; (iii) under extreme-fault injection (`retry_budget=0`), whether the kill-switch degrades honestly.

[Result]

gating 21 checks all pass: coverage tolerance ± 0.05 (fan ± 0.10), CPU fixed seed, all 21 gates pass (`gating.all_pass=true`, `gating.checked=21`). This means UDOS’s



uncertainty pipeline has, at all 21 checkpoints, no case of “claiming calibrated while actually not.”

contact gate: in P7’s embodied-control experiment, the contact gate lifts contact-related decisions from a 0% success rate under the direct mode (the pure semantic layer fails completely on contact tasks) to 100% under the motion/hybrid mode. This is an engineered abstention gate: when the task involves high-frequency contact, the pure semantic layer “knows what it does not know” and hands control to the motion prior. By the gate mean (`improvement_vs_v703.gate_mean_by_kind`), the gate-trigger rates per task kind are uniform 0.6341, accel 0.6624, collision 0.1544, spring 0.0—i.e., for the contact/spring tasks it is not good at, the gate correctly hands over control, while it keeps control for the uniform-motion tasks it is good at.

kill-switch: under the extreme fault `retry_budget=0`, the kill-switch degrades honestly (`accepted<24`, `governance.ok=false`) rather than faking success. This is metacognition at the system level—**the system honestly reports failure under extreme fault, rather than packaging failure as success**. P12 (evidence grading) institutionalizes this behavior: `verified/cpu-proto/cpu-proxy/unverified` three-level labels, plus `[CITATION NEEDED]/[RESULT NEEDED]` placeholders, write “not knowing” explicitly into the artifact.

[Validity threat] (i) The 0% → 100% contact gate comes from P7’s `cpu-proxy/cpu-proto` embodied demo, **not a real machine, not a large model**, and the contact dynamics are rule-based; (ii) gating `all_pass` is “the pipeline’s self-check all passed,” not “it does not panic under a real OOD”—it only verifies the pipeline’s internal self-consistency; (iii) all single seed, no multi-seed variance. See Section 9.2.

4.4 Evidence four: the evidence-grading system as institutionalized metacognition (P12 source)

[What we measure] We measure metacognition at the **organizational** (not single-model) level: whether an engineering system explicitly labels “how reliable each claim is” and is guarded by external tests.

[How we measure it] Source P12 (evidence-grading-driven reproducible agent systems engineering): check whether each quantitative claim is consistently labeled across code (`EvidenceGrade`), report (`evidence_grade` field), and documentation; check the “no



benchmark, no conclusion” discipline; check the single-source version and baseline-JSON regression gates.

[Result] P12 reports an engineering method that institutionalizes “not knowing”: each quantitative claim is consistently labeled verified/cpu-proto/unverified across code, report, and documentation; plus the “no benchmark, no conclusion” discipline; single-source version and baseline-JSON regression gates.

P12 also reports a real metacognitive failure point: from v7.4.3 to v7.4.10, eight consecutive versions stayed at v7.4.2 because the version number carried a “v” prefix not hit by the replacement regex, until v7.5.0 was exposed by a test. This “version-number incident” is an excellent metacognitive lesson—**the system thought it was tracking the version number, but it was not; its own metacognition was wrong.** This is exactly what F4 (metametacognitive calibration) handles.

This evidence must be read against the four-layer framework of Section 2.4. P12’s evidence-grading system essentially implements L3 (metacognition) at the organizational level: each number is tagged verified/cpu-proto/unverified, and the system “knows” how reliable each of its claims is. But the version-number incident shows that L3 itself can fail—the system thought it was tracking the version number (L3 metacognition), but the tracking logic had a bug, and its own metacognition about L3 was wrong. This is where L4 (metametacognition) comes in: **you need a second-order calibration to check whether your first-order calibration is really working.** P12’s “test guard” (using pytest tests to find the version-number incident) is a naive L4—it does not find the bug through the system’s “introspection” but through external tests verifying whether the system’s metacognitive claims are true. We list this finding as the direct motivation of F4: **explicit, externally auditable uncertainty channels (tests, gates, calibration sets) are more reliable than the system “saying it is calibrated.”**

[Validity threat] This is **process evidence, not cognitive-science evidence**—it proves that “institutional labeling works,” not that “the system truly understands itself”; the version-number incident itself precisely proves L3 can malfunction, and cannot be used as evidence that “L3 is already reliable.” See Section 9.3.



4.5 Evidence five: group-level metacognition (P2/P3/P11 source)

[What we measure] We measure metacognition at the **multi-agent group** level: whether a group system uses structural means (quorum, circuit breaker, accounting conservation) to “know what it does not know.”

[How we measure it] Source P2 (BFT-lite quorum), P3 (fault circuit-breaker governance), P11 (market settlement conservation), reading their acceptance numbers.

[Result] Metacognition is not only at the single-model level but also at the multi-agent group level. UDOS’s P2, P3, and P11 implement group-level “knowing what it does not know” from three angles:

- **P2 BFT-lite:** committee $n \geq 3f+1$, quorum $2f+1$. At $n=7, f=2$, all 24 work orders were accepted correctly. This is group-level “calibration”—it does not trust any single node’s judgment but uses quorum to take “majority agreement” as the trust estimate.
- **P3 fault circuit-breaker governance:** under clean/stressed runs, 24/24 closure; the fault fleet triggered 5 rework and isolated 3 faulty agents. This is group-level “abstention”—when group members show fault modes, the circuit breaker lets the group “stop” rather than continue confidently erring.
- **P11 market settlement conservation:** budget=paid=balance_sum=240, conserved=true. Conservation is a group-level “self-knowledge”—when the accounts do not balance, the system knows its internal bookkeeping is broken.

[Validity threat] (i) All three are single-seed cpu-*proto*; (ii) P3’s 24/24 closure is a single point estimate from two clean/stressed runs, and **whether F5 holds depends on ≥ 30 -seed replication** (Section 5, F5); (iii) “group-level metacognition” is this paper’s interpretive framework, not the original claim of P2/P3/P11. See Sections 9.1 and 9.5.

4.6 Evidence six: the world-model prediction-error channel (verified)

[What we measure] We measure the feasibility of the prediction residual as a natural OOD (“I do not know”) signal: does the world model’s prediction error for a new observation rise as it deviates from the training distribution.

[How we measure it] Data trace to the `improvement_vs_v701` and `latency_ms` sections of `v7_verification.json`. Compare the blind-test overall MSE of v7.0.1 and v7.0.3 on the same held-out test set; record the median single-step prediction latency.



[Result] The UDOS world model’s blind-test overall MSE dropped from 0.0459 in v7.0.1 to 0.01037 in v7.0.3 (about a 4.4-fold improvement); the median single-step prediction latency is 2.36 ms (`latency_ms.predict_next_batch1_median=2.3598` ms).

The meaning of this result for this paper is: **prediction error itself is an unknown detector**. When the world model’s prediction residual for an observation rises significantly, it often means that observation lies outside the model’s training distribution—i.e., where the model “does not know.” This is a direct corollary of the Ha & Schmidhuber (2018) framework on UDOS. We treat the prediction residual as a natural signal for OOD detection, but mark its limits: a rising residual does not equal “definitely OOD” (it could be observation noise), and a non-rising residual does not equal “definitely in-distribution” (OOD but the model confidently errs). The feasibility and limits of this channel are to be further tested by F4.

[Validity threat] (i) $0.0459 \rightarrow 0.01037$ is a **version-to-version self-comparison**, not a direct validation of OOD detection—it proves “the model gets more accurate as it trains,” not directly “the residual can reliably flag OOD”; (ii) single seed; (iii) the residual $\rightarrow$ OOD causal chain is not directly tested here and is a F4-pending inference. See Sections 9.4 and 9.5.

4.7 Summary: convergence of UDOS metacognitive evidence

Table 5. UDOS metacognitive evidence summary

#	Evidence	Metacognitive quantity	Key number	Grade
1	Split-conformal coverage	Calibration	80/90/95 all pass	verified
2	Acceleration a skill=-0.508	Identifiability self-awareness	Estimator worse than mean (0.9256 > 0.6138)	cpu-proto (main gate verified)
3	gating 21 / contact gate / kill-switch	Abstention	21 all_pass; 0% $\rightarrow$ 100%	verified (contact gate from P7)
4	Evidence-	Institutionalize	77 [CITATION	verified (P12)



#	Evidence	Metacognitive quantity	Key number	Grade
	grading three-level labels	d metacognition	NEEDED]	
5	BFT / circuit breaker / conservation	Group-level metacognition	24/24 acceptance; conservation 240	cpu-proto
6	Prediction-residual channel	OOD-detection feasibility	MSE 0.0459 → 0.010 37; latency 2.36 ms	verified

5. Five Falsifiable Predictions (F1–F5)

The value of a scientific proposition lies in what can kill it. Below, each of the five states its refutation condition and a **minimal decisive experiment**—the smallest experiment design that can judge its life or death.

F1 | Calibration is separable from performance

Statement: there exist training interventions or post-processing (e.g., temperature scaling, Platt scaling, conformal calibration) that improve calibration (Brier/ECE/AURC) without changing accuracy. If every means that improves “self-knowledge” improves accuracy in equal proportion, then calibration is not an independent axis.

UDOS seed: Guo et al. (2017) already proved temperature scaling improves calibration without changing accuracy; UDOS’s split conformal is also post-processing, not retraining.

Refutation condition: if, under a strict contrast, every calibration-only intervention improves accuracy in equal proportion (i.e., calibration improvement is always a by-product of performance improvement), then our “calibration is an independent axis” is falsified, retreating to “calibration is only an accessory metric of performance.”



Minimal decisive experiment: take a single trained checkpoint (no retraining needed), run a one-dimensional grid search over temperature T on an independent calibration set, freeze the network weights, and measure held-out ECE/Brier and top-1/top-5 accuracy with 5-fold cross-validation. **Criterion:** if some T makes ECE drop significantly (e.g., relative drop $\geq 30\%$) while accuracy changes within statistical noise (e.g., $< 0.1\%$), F1 survives; if every measurable drop in ECE is accompanied by a proportional rise in accuracy, F1 dies. This is a minimal experiment with a single checkpoint, a single hyperparameter, and a single data split.

F2 | The identifiability boundary: when the sensing structure is insufficient, adding parameters does not help

Statement: when the sensing structure is insufficient to identify a latent parameter (e.g., estimating acceleration from a 6-frame window), no scale of model ($\text{skill} \leq 0$) can identify it; enlarging model parameters or adding training data does not make skill rebound.

UDOS seed: acceleration a $\text{skill} = -0.508$ (6-frame window; estimator MAE 0.9256 > mean 0.6138).

Refutation condition: if, under a 6-frame window, enlarging model parameters by 5–10 $\times$ makes acceleration skill stably rebound to positive (i.e., a large model, by capacity, “hard-learns” the acceleration information not present in the sensing structure), then our “identifiability is a structural property, not a parameter property” is falsified—it means a large model can break through the hard boundary of the sensing structure by capacity. This is our most radical and most easily tested prediction at scale.

Minimal decisive experiment: hold the observation window fixed at 6 frames, and scale only the estimator capacity (GRU hidden dimension from 256 to 512, 1024; layers $2 \rightarrow 4$), train to convergence on the same synthetic data, and measure acceleration a skill. **Criterion:** if skill remains ≤ 0 across three capacities (or does not rise with capacity), F2 survives; if skill grows stably past 0 to significantly positive (e.g., > 0.2) with capacity, F2 dies. **Key control:** another arm lengthens the window from 6 to 12/24 frames as a positive control—if skill immediately turns positive when the window lengthens but not when capacity grows, it cleanly separates the two variables “sensing structure” and “model capacity.” This is exactly the judgment we most want to see.



F3 | Risk-coverage dominance: conformal/abstention gates achieve higher coverage at the same error-rate budget

Statement: a system with conformal calibration or an explicit abstention gate achieves, at a given error-rate budget, significantly higher coverage (fraction of samples answered) than a gated system (smaller AURC).

UDOS seed: UDOS’s gating all 21 pass; the contact gate lifts success from 0% to 100%.

Refutation condition: if, on a large-scale benchmark, a gated and a gated-out system have no significant difference in coverage at the same error-rate budget (the AURC difference is within statistical noise), then “abstention is valuable” is falsified, retreating to “abstention just pushes hard problems to others without solving them.”

Minimal decisive experiment: on a single benchmark dataset, set two arms—(A) bare model, (B) the same bare model + conformal/confidence gate; fix the same target error-rate budget, sweep the threshold, and measure the fraction of samples each can cover (answer). **Criterion:** at the same error rate, if B’s coverage is significantly higher than A (AURC significantly smaller), F3 survives; if the two arms’ risk-coverage curves overlap statistically (AURC difference within noise), F3 dies. A single dataset, two arms, and a threshold sweep suffice.

F4 | Metacognitive auditability: explicit uncertainty channels degrade slower under OOD than verbalized confidence

Statement: explicit uncertainty channels (conformal intervals, parameter fan, prediction residual) degrade slower under out-of-distribution data than verbalized confidence (the model directly saying “I am X% confident”) in ECE.

UDOS seed: UDOS’s blind path passes all three levels 80/90/95; the prediction-residual channel (MSE 0.0459 → 0.01037) has verified feasibility as an OOD signal.

Refutation condition: if, on an OOD benchmark, explicit uncertainty channels and verbalized confidence show no significant difference in ECE degradation (or verbalized is more stable), then “explicit channels are superior to verbalized” is falsified—it means the confidence the model directly states is already reliable enough, and no dedicated uncertainty pipeline is needed.

Minimal decisive experiment: construct a known OOD shift (e.g., train on uniform-motion dynamics, test on spring/collision dynamics), gradually increase the fraction of



OOD samples in the test set; for each fraction, simultaneously measure two arms' ECE—(A) the coverage error corresponding to the conformal interval bandwidth (explicit channel), (B) the difference between the model's self-reported confidence and its actual accuracy (verbalized confidence). **Criterion:** as the OOD fraction rises, if A's ECE degrades significantly slower than B (A stays near nominal coverage at high OOD fractions while B's self-reported confidence has already severely deviated), F4 survives; if the two curves overlap or B is more stable, F4 dies. This is the only experiment that can directly collide “the measured” against “the said” on the same OOD axis.

F5 | Group-level metacognition: systems with a Trace/Stop matrix have higher closure rates under stress injection

Statement: a multi-agent system equipped with an Owner/Trace/Stop Condition matrix and BFT quorum has, under fault stress injection, a closure rate (fraction that either completes correctly or circuits-breaks correctly) significantly higher than a system without the matrix.

UDOS seed: P3 already measured clean/stressed two runs at 24/24 closure; but **this is single seed and requires ≥ 30 seeds for significance.**

Refutation condition: if rerunning with ≥ 30 seeds, the closure-rate difference between the matrix and no-matrix systems under stress injection has a 95% confidence interval containing 0, then F5 is falsified—it means that one 24/24 was noise rather than a structural effect. We explicitly admit: this is currently only a point estimate, and whether F5 holds depends on multi-seed replication.

Minimal decisive experiment: run P3's clean/stressed experiment on two arms (with-matrix vs no-matrix), each for ≥ 30 seeds, measure the closure rate under stress injection, and report the 95% confidence interval of the closure-rate difference. **Criterion:** if the 95% CI lies entirely outside 0 (with-matrix higher), F5 survives; if the CI contains 0, F5 dies. The minimal cost of this experiment is to run the existing single-seed script 30 times, with no new code—it is the cheapest of the five, and the one we most want to do next.

6. Objections and Boundaries: What This Paper Does Not Claim

The lifespan of a flagship paper depends on whether it fully writes its objections. This chapter lists, one by one, serious objections to the central claim.



6.1 Calibration ≠ consciousness: “I do not know” can be trained as a behavior pattern

The most direct objection comes from behaviorism: **a system can be trained to output “I do not know” at low confidence, but this does not mean it has any sense of self.** Temperature scaling only divides the logits by a scalar; conformal prediction only takes quantiles on a calibration set; the abstention gate only compares a confidence to a threshold. All of these are mechanical mathematical operations, none requiring “the system knows what it is doing.” Is it not the case that calling these operations “metacognition” is merely slapping a psychology label on a mechanical process?

Let us ground this objection at the mechanism level rather than leave it as a slogan. All the math of temperature scaling is $p_i = \text{softmax}(z_i/T)$ —a scalar T dividing the logits, not changing the relative class order, only globally compressing/stretching confidence. All the math of conformal prediction is computing non-conformity scores on the calibration set and taking the $1 - \alpha$ quantile as the bandwidth—it is a lookup action, not requiring the model to “understand” what the bandwidth means. All the math of the abstention gate is $1[s(x) > \tau]$ —comparing a score $s(x)$ to a threshold τ and outputting 0 or 1. Putting these three together, **not a single step requires the system “to reflect on itself reflecting.”** A process that can be completed by pure forward matrix multiplication, when called “metacognition,” is indeed suspicious on its face.

Our response is twofold. First, we **fully agree** with this objection and write it into the title and abstract: this paper does not claim consciousness, only three measurable quantities. “Metacognition” is an engineering term here, referring to the three quantities “calibration + abstention + identifiability self-awareness,” not Flavell’s metacognitive experience. Second, our stance is precisely behaviorist: **whether a system “has self-knowledge,” we make no metaphysical judgment; we only look at whether it is behaviorally calibrated, whether it abstains, and whether it knows the identifiability boundary.** If it does these three things behaviorally, we say it “has engineering self-knowledge”; whether it has some subjective experience internally, this paper does not answer—that is Section 7. We replace “truly has self-knowledge” with “has engineering self-knowledge,” and this replacement itself is the acceptance of the behaviorist objection.

It must be stated further: **all our engineering evidence (coverage, skill, gating) proves only behavior patterns, not any internal state.** A system calibrated at 0.7839/0.8984/0.9577 can entirely achieve this “without any experience”—because split



conformal is, by design, a parameter-free lookup. We do not think this weakens the paper; on the contrary, it is our stance: **to reduce “self-knowledge” from the metaphysical question of “inner experience” to the engineering question of “whether the behavior is calibrated/abstains/knows the boundary” is the precondition for being able to build it.** If we claimed “the system has consciousness,” the mechanical-operation argument above would be fatal; but we claim only “the system’s behavior satisfies three engineering necessary conditions,” and the mechanical operations precisely prove that these three conditions can be engineered.

6.2 The moral hazard of abstention: over-abstention = evading the task, and confidence can be adversarially deceived

A serious engineering objection is: **over-abstention equals evading the task.** If a system abstains whenever it is slightly uncertain, it loses its value as a system—the user wants answers, not a string of “I do not know.” In high-risk scenarios such as medicine, law, and engineering, “I do not know” is sometimes more dangerous than “wrong answer” (because it pushes responsibility onto the user).

Another objection is: **confidence can be adversarially deceived.** A carefully constructed adversarial sample can make the model output extremely high confidence for a completely wrong prediction; conversely, an ordinary but noise-perturbed input can make the model output very low confidence for a correct prediction. The “exchangeability” premise of conformal prediction fails under OOD—when the test distribution differs from the calibration distribution, the coverage guarantee no longer holds.

We accept these objections and make them part of our boundary. Our stance is: **abstention is not the more the better, but must match the task risk.** The whole point of the risk–coverage curve is to let the user make an explicit trade-off between “coverage” and “error rate”—it is not a binary judgment of “should I abstain,” but a continuous decision of “at the current risk budget, how far should I abstain.” The failure of conformal coverage under OOD is the research object of our F4, not a limitation we pretend does not exist.

The “moral hazard of abstention” must be made more concrete, because in high-risk scenarios it is not an abstract philosophical question. In a medical-diagnosis scenario, a system saying “this I do not know” may push the patient to a doctor—good when doctors are plentiful, but when doctors are scarce, “I do not know” is itself a dereliction of duty. In



an autonomous-driving scenario, a system returns control to the human driver when uncertain, but if the human driver is already distracted (which is precisely why autonomous driving is needed), “handing back” is more dangerous than “the system handles it itself.” Our stance is: **abstention is not a free safety action; it has its own cost—a cost that must be explicitly modeled into the risk–coverage curve.** A good risk–coverage curve is not just “how many answered vs how many wrong,” but should also include “how many rejected vs how much the rejection cost is.” Our current AURC measures only the former half; the latter half (rejection cost) is the next step to fill—an honest limitation of this paper.

6.3 Unreliability of introspection: LLM verbalized confidence disconnects from true correctness

Kadavath et al. (2022) found that an LLM’s internal confidence correlates with correctness but is far from perfect; Reflexion’s verbalized reflection can also reflect wrongly. A deeper objection is: **letting a system directly say “how certain am I” is itself unreliable—because the system’s own metacognition itself needs calibration.** This is L4 (metametacognition) in Table 1 of Section 2.4: the recalibration of calibration.

Let us write the “disconnect” as testable evidence rather than vague suspicion. Among Kadavath et al.’s (2022) findings, three points directly support “verbalized confidence will disconnect”: first, the model’s internal signal **correlates with correctness but is far from perfect**—it is not a reliable readout table; second, **RLHF fine-tuning degrades calibration**—i.e., a model optimized to “sound more human” may have its self-reported confidence trained to be more confident rather than more accurate; third, the model can “non-verbally” tell right from wrong (choose correctly in multiple-choice) but **may not translate this discrimination into numerical confidence**—i.e., there is a real, measurable gap between “knowing whether you are right” and “being able to say how certain you are.” This is exactly what F4 wants to collide with a minimal decisive experiment: **if verbalized confidence truly disconnects from correctness, then under OOD it will degrade faster than the explicit channel measured on an independent calibration set.**

Reflexion’s verbalized reflection has an additional disconnect source: **it may reflect wrongly, or form a wrong attribution of its own failure.** An agent that writes “where did I go wrong this time” into text may well attribute the cause wrongly in the text—and



this wrong attribution is written into episodic memory, contaminating the next trial. The verbalized channel turns “where did I err” into a text-generation problem, and text generation itself can hallucinate.

We accept this objection. This is precisely why our F4 distinguishes “explicit uncertainty channels” from “verbalized confidence”: explicit channels (conformal intervals, parameter fan, prediction residual) are measured on an independent calibration set and do not depend on the model “saying how certain it is”; verbalized confidence is what the model says about itself, and it itself needs calibration. Our stance is: **do not believe what the model says; look at what it measures**. An interval that meets coverage on an independent calibration set is more reliable than the model directly saying “I am 90% confident.” This is also why we treat Kadavath’s finding as both an optimistic face (the model already has a signal inside) and a pessimistic face (this signal is far from perfect and is further degraded by RLHF)—**the correct engineering action is to treat this present-but-unreliable internal signal as candidate confidence, then calibrate it into a reliable quantity with an independent calibration set, rather than directly trusting its self-report**.

6.4 The limits of cultural propositions: the Analects is a normative ideal, not an empirical theory

An objection that must be faced honestly is: **the Analects’ “knowing what one knows” is a normative ideal (should), not an empirical theory (is)**. It tells you how one *ought* to act, not whether a system *actually can*. Is treating a 2,500-year-old moral exhortation as the intellectual origin of an engineering proposition not the logical fallacy of “appeal to authority”? The Delphic “know thyself” is the same—it is an inscription, not a falsifiable scientific hypothesis.

We accept this objection and write it into the methodological discipline of Section 1.4: **the Analects and Delphi serve only as epigraphs, not as scientific evidence**. All our scientific claims rest on the three measurable quantities of Section 2 and the UDOS measurements of Section 4, not on the authority of Confucius. We cite Confucius to clarify what ancient intuition this paper wants to engineer—not to use Confucius’s words to prove this paper right. **If our F1–F5 are overturned by experiment, Confucius’s words cannot save this paper**. This discipline itself is the expression of “knowing what one knows” in writing: knowing which things are evidence and which are not.



The distinction between “normative ideal” and “empirical theory” must be explained more thoroughly, because it is the core of our writing discipline. The Analects’ “knowing what one knows” is a normative statement—it tells a person they ought to face their knowledge boundary honestly. It is not a descriptive statement—it does not tell you whether humans actually can, under what conditions they can, or what happens after they do. Our work is to turn this normative statement into a descriptive research program: **we do not assume the system “ought” to have self-knowledge, but measure whether it “actually” has it, under what conditions, and how to make it have it.** Confucius gives the direction (one ought to be honest), but we turn the direction into falsifiable propositions (is it calibrated, is the abstention accurate, where is the identifiability boundary). This conversion itself is the scientific step “from norm to empirics”—it does not deny Confucius but makes Confucius’s intuition scientific. The Delphic “know thyself” is the same: it is an inscription-like call; decomposing it into three measurable quantities is doing the same conversion.

6.5 cpu-proto external validity: synthetic dynamics / simulated agents cannot assert real LLMs / robots

This is the landing point of our academic honesty. All UDOS measurements: (a) are a Linux CPU deterministic prototype, not a real LLM/real machine; (b) the semantic judge is rule-based, not a VLM; (c) the environment is a point-mass double-integrator agent, not MuJoCo-MJX contact simulation or a real machine; (d) all are single-seed pilot point estimates with no 95% confidence interval. The acceleration a negative result is measured on a synthetically injected defect structure, and real-world “insufficient observation” is more complex; the conformal coverage is measured on a synthetic-dynamics world, and the calibration problem of real text distributions is far more complex.

Therefore, all our UDOS numbers are only “silicon preliminary evidence,” not “a law verified on a real LLM.” We position ourselves as: a research program + silicon preliminary evidence, proposing three measurable quantities and five falsifiable predictions—not an already-verified “theory of self-knowledge.” F2 already writes the retreat: if a large model can break through the sensing-structure hard boundary by capacity, this paper retreats to “the sensing-structure constraint of small models.”



7. Boundary with the Consciousness Proposition (a separate section, declaring out of scope)

This section separately handles consciousness-philosophy propositions and explicitly declares: **the following discussion is outside the scientific scope of this paper.**

This paper nowhere claims “a system is conscious (consciousness).” The three quantities we measure—calibration, abstention, identifiability self-awareness—are all behavioral/engineering quantities, involving no judgment about subjective experience (qualia), sense of self, or phenomenal consciousness. A system that is behaviorally calibrated, abstains, and knows the identifiability boundary is a wholly different question from whether it “internally has some experience.”

In the consciousness-philosophy literature, Graziano’s AST (Attention Schema Theory) and Global Workspace Theory (proposed by Baars, developed by Dehaene) are two widely discussed hypotheses. AST claims that consciousness is the brain’s simplified model of its own attention process; GWT claims that consciousness is the state in which information enters the global workspace and is broadcast broadly to multiple downstream modules. Both link “consciousness” to some “self-model” or “global accessibility,” and thus have a surface resemblance to our M_{self} .

But we do not endorse, cite, or test these two hypotheses. There are three reasons: first, AST and GWT are both scientific hypotheses about human consciousness, and their applicability to silicon systems is itself disputed; second, our M_{self} is an engineering term (a numerical estimate of one’s own ability), not the same concept as AST’s “attention schema”; third, linking the calibration behavior of an engineering system to human consciousness is the category error we are most wary of—a calibrated system need not be conscious, and a conscious system need not be calibrated.

The existence of this section is itself the expression of our “knowing what one knows” discipline: **we know we do not know what consciousness is, and we do not pretend to.** This paper only does what it can—make the three engineering necessary conditions of “self-knowledge” into measurable, falsifiable quantities. Whether these three quantities suffice to constitute “consciousness,” this paper does not answer; we leave that question to philosophy of mind and cognitive science, which are their specialty, not ours.



8. Discussion

8.1 Implications for architecture design: install the brakes before pressing the accelerator

If our framework holds, then an architect’s decision order should be: **before making the model’s capability large, first install its brakes**. A large model without calibration, without abstention, and unaware of the identifiability boundary is not a more powerful model but a more dangerous one—the stronger its in-distribution capability, the more severe the consequence of confidently lying out of distribution.

This principle has three engineering corollaries. First, **calibration should be a first-class acceptance metric, not a post-hoc patch after training**. Guo et al. (2017) prove temperature scaling is post-processing, but that does not mean calibration can be ignored—it means calibration can be cheaply improved, provided you measure it. UDOS’s gating all 21 pass is precisely making “is it calibrated” a CI gate. Second, **abstention should match task risk, not be a binary switch**. The whole value of the risk–coverage curve is to let the user make a continuous trade-off under a risk budget. Third, **identifiability self-awareness should be considered at the sensing-structure design stage**. P6’s acceleration negative result reminds us: if your observation window is too short, no matter how strong a model you use, it is useless—the correct engineering action is to lengthen the observation window, not to swap in a larger model.

8.2 An engineeringized prescription for the “hallucination” problem

The engineering essence of hallucination is precisely “answering confidently where one should not.” Our stance is: **hallucination is not a bug to be “eliminated” but a boundary to be “managed.”** You cannot make a system answer correctly on 100% of samples; what you can do is make it, where it does not know, (a) output a wide interval, (b) abstain, or (c) turn outward to external knowledge (RAG/Toolformer). UDOS’s three pieces of evidence—conformal coverage, the acceleration negative result, and gating all_pass—correspond respectively to these three strategies.

This contrasts with the industry’s current approach of “eliminating hallucination with larger models.” Our stance is: **a larger model can improve in-distribution capability but cannot eliminate out-of-distribution hallucination**. Because the root of hallucination is not “the model is not large enough” but “the model does not know what it



does not know”—this is a self-boundary problem, not a scale problem. F2 already writes this stance as a falsifiable prediction: if a large model can break through the sensing-structure hard boundary by capacity, this paper retreats to “the sensing-structure constraint of small models.”

The engineering definition of “hallucination” must be tightened once more, because it is this paper’s core problem awareness. By hallucination we do not mean “the model said a factually wrong thing” (because any finite model can be wrong), but rather **“the model, about something it essentially does not know, outputs an answer as confident as the correct one.”** The distinction is crucial: the former is “insufficient capability,” the latter is “missing self-boundary.” A calibrated system should output a wide interval or abstain when capability is insufficient; an uncalibrated system still outputs a confident point estimate when capability is insufficient—that is the dangerous hallucination. UDOS’s acceleration a negative result is a miniature “hallucination”: the estimator, for a parameter it essentially cannot estimate, outputs a point estimate worse than the mean, without realizing it cannot estimate it. **The “knowing what one knows” we want to engineer is precisely to make the system, in such cases, no longer confidently answer hard.**

8.3 Relation to P0/P0b: three orthogonal axes, not three alternatives

We do not negate P0 and P0b but supplement the third axis they lack. The core statement of the three orthogonal axes is: **structure determines what F looks like, scale determines how far along F you travel, and self-boundaries determine the point at which you must stop.** The three are multiplied, not added. A system with correct structure and sufficient scale but no brakes is still dangerous; a system with sensitive brakes but wrong structure and insufficient scale is still a toy.

This triad also has a restrained implication for the discussion of “AGI.” This paper does not predict when AGI arrives, nor claim that self-boundaries alone lead to AGI. It offers only a research stance: **if AGI is “a system that knows what it knows and what it does not know,” then the third axis is not the icing on the cake of AGI but its necessary condition.** A super-large system without a self-boundary is not AGI but a super-large confident liar.

The distinction between the “triad” and a “multiplicative relation” must be clarified, because it is an easily misread point of our framework. P0 uses a multiplicative model of “structure $\times$ dynamics $\times$ energy” (if any is zero, capability is zero); we write “structure $\times$



scale $\times$ self-boundaries” as three orthogonal axes, but the relation among the three is not strictly multiplicative—more precisely, **self-boundaries are not a “source of capability” alongside structure and scale, but a “constraint condition.”** Structure and scale determine how much the system can do (the capability upper bound); self-boundaries determine how much it dares to do and where it stops (the boundary of capability use). A system with a very high capability upper bound but a missing self-boundary is not “capability multiplied by zero” but “capability losing control out of distribution”—it is extremely strong in-distribution and extremely wrong out of distribution. This is why we repeatedly use “brakes” rather than “engine” as the metaphor for the third axis: brakes do not directly produce power, but the stronger the power of a car without brakes, the more dangerous it is.

The relation of the triad to P0b’s “three-segment curve” should also be pointed out. P0b proposes that capability vs scale follows “local power law + phase-transition breakpoint + saturation/negative return” in three segments. Our third axis can be read as: **self-boundaries determine the system’s “safe working region” on the three-segment curve.** In segment 1 (local power law, in-distribution), the system can answer confidently; in segment 2 (phase-transition breakpoint, distribution boundary), it should begin to abstain; in segment 3 (saturation/negative return, out-of-distribution), it should stop entirely or turn outward. This reading connects P0b’s “curve shape” with P0c’s “abstention point”: **not every scale segment suits confident answering; the self-boundary is the signal that tells the system “which segment you are on now.”**

8.4 Limitations and next steps

This paper must honestly list how far it still is from a “law.” First, **multi-seed replication**: all UDOS are single-seed pilot point estimates, and whether F2/F5 hold depends on multi-seed replication. Second, **real-machine/GPU gate**: the UDOS conformal coverage is measured on a synthetic-dynamics world, and the calibration problem of real text distributions is far more complex; moving split conformal to the output space of a real LLM is the next step. Third, **L4 metacognition layer**: our four-layer framework only does L3, and L4 (recalibration of calibration) is an open layer—whether a system that knows how well it is calibrated itself needs a second-order calibration is the research object of F4. Fourth, **the boundary with consciousness philosophy**: we have declared



out of scope in Section 7, but the surface resemblance between M_{self} and AST/GWT requires a dedicated discussion in philosophy of mind, which is beyond this paper’s scope.

9. Threats to Validity

This is this paper’s independent threats-to-validity chapter. Chapter 6 handles “objection positions” (calibration $\neq$ consciousness, the moral hazard of abstention, unreliability of introspection, the limits of cultural propositions); this chapter stands in the reviewer’s position to list the engineering evidence’s threats by internal/external/construct/statistical/causal type. **This chapter still holds the red line: the paper nowhere claims the system is conscious, and reports only behavioral/engineering quantities.**

9.1 Internal validity: is the oracle/blind contrast really caused by “whether parameters are known”

- **Contrast purity:** oracle and blind use the same world model, the same calibration set, and the same split-conformal pipeline; the only difference is whether ground-truth hidden physical parameters are injected. This contrast design is clean, but a third arm beyond oracle/blind (randomly injecting wrong parameters) was not tested.
- **The negative result is an injected defect structure:** acceleration skill= -0.508 is measured on a synthetically injected 6-frame short-window structure ($n=160$); real-world “insufficient observation” is more complex and cannot be directly extrapolated.
- **Single seed:** all numbers are fixed-seed, with no 95% confidence interval. **≥ 30 seeds are required before submission.**

9.2 External validity: synthetic dynamics is not the real LLM output space

UDOS is a Linux CPU deterministic prototype; the world model is a point-mass double-integrator agent (967,796 parameters); the conformal coverage is measured on a synthetic-dynamics world. The calibration of real text distributions and real LLM hallucination are far more complex. We do not claim the split-conformal coverage transfers directly to the real LLM output space.



9.3 Construct validity: calibration $\neq$ consciousness; ECE and AURC measure different things

Our “metacognition” is an engineering term (calibration + abstention + identifiability self-awareness), not Flavell’s metacognitive experience. We must guard against: a system that is behaviorally calibrated and abstains does not mean it “has self-knowledge”; temperature scaling/conformal/gating are all mechanical mathematical operations (Section 6.1). ECE measures absolute confidence and AURC measures confidence ranking; the two must not be conflated, and reporting both is precisely for this reason. The distinction “high observability (0.798) $\neq$ identifiable” (Section 4.2) is also a construct self-guard.

9.4 Statistical validity: premises and limits of the conformal guarantee

Split conformal’s marginal coverage guarantee relies on “calibration set exchangeable with test set”—the premise fails under distribution shift / OOD, and coverage degrades (discussed in P5 Section 7). We report marginal coverage, not conditional coverage; the fan ± 0.10 tolerance is wide. Single-point small-sample results (e.g., 16-sample sampling of the AIME style) must not be written as conclusions.

9.5 Causal validity: explicit channels vs verbalized confidence

F4 claims “explicit uncertainty channels degrade slower under OOD,” but currently there is only the pilot of the blind path passing all three levels, with no large-scale OOD contrast against verbalized confidence. This is F4 pending, not something this paper has proven.

9.6 Summary of validity threats

Threat type	Most likely attack	How we defend	Open ledger
Internal	Negative result is injected defect, not the real world	Synthetic structure labeled, $n=160$ noted	Re-test under real insufficient observation
External	Synthetic dynamics $\neq$ real LLM	Four evidence grades, no real-machine claim	split conformal on a real LLM
Construct	Calibration misread as consciousness	Paper nowhere claims consciousness, Section 7 out-of-	—



Threat type	Most likely attack	How we defend	Open ledger
		scope declaration	
Statistical	Conformal exchangeability fails under OOD	Marginal $\neq$ conditional coverage declared	Measure OOD coverage degradation
Causal	Explicit channels need not beat verbal	F4 left as a falsifiable prediction	Large-scale OOD contrast

10. Conclusion

P0 proved that structure is a substrate-invariant; P0b added scale; this paper adds the third axis both froze—the self-boundary. We replace the unmeasurable word “wisdom” with its three engineering necessary conditions: **calibration** (confidence consistent with accuracy), **abstention** (actively abstaining on high-risk samples), and **identifiability self-awareness** (knowing which latent variables are unknowable under the current sensing structure). Together these are the metacognitive triple, with the risk–coverage curve and AURC as the primary metric.

On a 967,796-parameter world model, UDOS yields three direct pieces of evidence: split conformal passes all three levels 80/90/95; acceleration $\mathbf{a}$ under a 6-frame window has $\text{skill} = -0.508$ (the estimator is worse than the mean, and the correct action is abstention); gating all 21 checks pass. These are all silicon preliminary evidence, not a law verified on a real LLM.

A meta-stance must be pointed out in the conclusion: **this paper does not pursue “proving the system has wisdom,” but “giving the engineering necessary conditions of wisdom.”** Wisdom itself is an unmeasurable word; we replace it with the three measurable quantities—calibration, abstention, identifiability self-awareness—and argue that until these three are made measurable, acceptable, and falsifiable, any claim that “the system has wisdom” is unsound. This is a much weaker but also much harder claim than “proving wisdom”: it does not answer “what is wisdom,” only “before claiming wisdom, the three measurable things the system must first do.”



This paper does not overstate its case. Chapter 6 fully acknowledges: calibration $\neq$ consciousness, abstention carries moral hazard, verbalized confidence disconnects from true correctness, cultural propositions have limits, and cpu-*proto* external validity is limited. Section 7 separately declares: this paper does not claim consciousness, and AST and GWT are placed only as hypotheses, not as conclusions.

Finally, the relation of this paper to P0/P0b is closed into one memorable sentence. P0 says: **space structure is itself intelligence**. P0b says: **structure + scale is itself intelligence**. This paper says: **knowing what one knows is itself the third axis of intelligence**. Together, the three sentences form a complete engineering judgment: a system's maturity lies not in how much it knows (scale), nor in how it organizes what it knows (structure), but in whether it knows what it knows and what it does not know (the self-boundary). The first two make the system powerful; the third makes it trustworthy. A powerful but untrustworthy system is not a better system but a more dangerous one. This is the entire intent of this paper as the prologue of the flagship trilogy: **not to announce a new truth, but to turn the “brakes” from a metaphor into a measurable, acceptable, falsifiable engineering object**.

Returning to that poetic line: **the coming-of-age of AI lies not in how much it knows, but in how well it knows what it does not know**. The Analects has said this for two and a half millennia; what this paper does is rebuild it from a moral exhortation into a falsifiable engineering proposition—calibration, abstention, identifiability self-awareness. Until these three are made measurable, acceptable, and falsifiable, talking about “the system has wisdom” is unsound. This is the entire intent of this paper as the flagship trilogy's prologue: not to announce a new truth, but to set up a target that experiment can overturn.

References

Note: AI/systems literature is reused from our volume's *Reference Library* (verified 2026-09-19,); this paper does not rewrite the library itself. Below are the entries newly verified by this paper, entry-by-entry on 2026-09-20. Anything unverifiable is marked [CITATION NEEDED].



B. Newly verified by this paper (2026-09-20)

1. **Flavell, J. H.** (1979). *Metacognition and Cognitive Monitoring: A New Area of Cognitive-Developmental Inquiry*. **American Psychologist**, 34(10), 906–911. DOI:10.1037/0003-066X.34.10.906.
2. **Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q.** (2017). *On Calibration of Modern Neural Networks*. **ICML 2017**, PMLR 70:1321–1330. arXiv:1706.04599.
3. **Chow, C. K.** (1957). *An optimum character recognition system using decision functions*. **IRE Transactions on Electronic Computers**, EC-6, 247–254.
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5. **El-Yaniv, R. & Wiener, Y.** (2010). *On the foundations of noise-free selective classification*. **JMLR**, 11, 1605–1641.
6. **Geifman, Y. & El-Yaniv, R.** (2017). *Selective Classification for Deep Neural Networks*. **NeurIPS 2017**.
7. **Geifman, Y., Uziel, G., & El-Yaniv, R.** (2019). *SelectiveNet: A Deep Neural Network with an Integrated Reject Option*. arXiv:1901.09192.
8. **Kadavath, S., Conerly, T., Askell, A., et al.** (2022). *Language Models (Mostly) Know What They Know*. arXiv:2207.05221.
9. **Shinn, N., Cassano, F., Labash, B., Gopinath, A., Narasimhan, K., & Yao, S.** (2023). *Reflexion: Language Agents with Verbal Reinforcement Learning*. **NeurIPS 2023**. arXiv:2303.11366.
10. **Ha, D. & Schmidhuber, J.** (2018). *World Models*. arXiv:1803.10122.
11. **Hendrycks, D. & Gimpel, K.** (2017). *A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks*. **ICLR 2017**. arXiv:1610.02136. [exact abs page [CITATION NEEDED]]
12. **Ji, Z., Lee, N., Frieske, R., et al.** (2023). *Survey of Hallucination in Natural Language Generation*. **ACM Computing Surveys**. [exact volume/pages [CITATION NEEDED]]
13. **Bellman, R. & Åström, K. J.** (1970). *On structural identifiability*. **Mathematical Biosciences**. [exact volume/pages [CITATION NEEDED]]



C. Reused from the Reference Library (2026-09-19)

14. Angelopoulos, A. N. & Bates, S. (2021). *A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification*. arXiv:2107.07511.
15. Lewis, P. et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS 2020. arXiv:2005.11401.
16. Schick, T. et al. (2023). *Toolformer: Language Models Can Teach Themselves to Use Tools*. NeurIPS 2023. arXiv:2302.04761.
17. Vaswani, A. et al. (2017). *Attention Is All You Need*. NeurIPS 2017. arXiv:1706.03762.
18. Hoffmann, J. et al. (2022). *Training Compute-Optimal Large Language Models (Chinchilla)*. arXiv:2203.15556.
19. Kaplan, J. et al. (2020). *Scaling Laws for Neural Language Models*. arXiv:2001.08361.
20. Castro, M. & Liskov, B. (1999). *Practical Byzantine Fault Tolerance (PBFT)*. OSDI 1999.

D. Unverified placeholders (do not use as real citations)

- [CITATION NEEDED]: Chow 1970 exact volume/pages; Hendrycks & Gimpel 2017 abs page; Ji et al. 2023 ACM CSUR exact volume/pages; Bellman & Åström 1970 exact volume/pages; the exact literature for Graziano AST and Global Workspace Theory (this paper places them only as hypotheses, not as concluding citations).

Appendix A. UDOS measurement reproduction index (numbers traced to reports7)

Number in this paper	Source file	Field	Grade
Conformal 80/90/95 oracle	v7_verification.json	coverage.oracle_alpha_*	verified
0.7839/0.8984/0.9577; blind		coverage.blind_alpha_*	



Number in this paper	Source file	Field	Grade
0.7758/0.9001/0.95 59			
Bandwidth oracle 0.043/0.0858/0.161 2; blind 0.0504/0.1081/0.20 52	v7_verification. n.json	*.bandwidth_ma rginal	verified
Acceleration a skill=-0.508 (MAE 0.9256 vs mean 0.6138, observability 0.798); v0 0.318 (observability 0.771); ω 0.426; v2 0.82	v7_verification. n.json	estimator_para m_error.*	cpu-proto (main gate verified)
gating 21 all_pass; tolerance ± 0.05 ; gate mean uniform 0.6341/accel 0.6624/collision 0.1544/spring 0.0	v7_verification. n.json	gating / improvement_vs _v703.gate_mea n_by_kind	verified
World model 967,796 params; blind overall 0.01037 vs v701 0.0459	v7_verification. n.json	params / improvement_vs _v701	verified
Single-step latency 2.36 ms	v7_verification. n.json	latency_ms.pre dict_next_batch h1_median	verified



Number in this paper	Source file	Field	Grade
Contact gate 0% → 100%; kill- switch retry_budget=0	P7 source + P12	embodied_hybri d_demo / governance	cpu-proto

Appendix B. Self-check of consistency with P0/P0b/P5/P6/P12

- All UDOS numbers are consistent with P0, P0b, P5, P6, P12; the coverage range is corrected per the 99 review to the six-group merged truth values (oracle 0.7839–0.9577, blind 0.7758–0.9559).
- P0’s “structure is the fourth axis” is extended here to “three orthogonal axes”: structure fixes the functional form, scale fixes the position, self-boundaries fix the abstention point; the three do not conflict and are orthogonal complements.
- P5’s conformal prediction and P6’s identifiability are re-framed here as two quantities of the “metacognitive triple” (calibration, identifiability self-awareness), with consistent numbers and caliber.
- P12’s evidence-grading system is cited here as “institutionalized metacognition,” consistent with the three-level labels and the [CITATION NEEDED]/[RESULT NEEDED] placeholder caliber.
- CTM in this repository is the Continuous Thought Machine, not a “causal engine”; causal structure belongs to the `causal_parents` field of PCE-Format, and this paper does not misattribute CTM as causal reasoning.
- This paper does not modify the *Reference Library* itself; the newly verified entries (Section B, 13 items) are listed only here, pending unified merge into the library.
- All AGI/ASI/twin-earth Lv1–7 discussion in this paper is vision-only and labeled unverified; specific hardware names are unverified and not written into the body.

(End. P0c is the prologue of the flagship trilogy; together with P0 and P0b it forms the “structure × scale × self-boundaries” three-orthogonal-axes framework. All UDOS numbers trace to `release-v5.4.4/udos-engine/reports7/`; evidence grades and reproduction steps are in Appendix A.)